

sect113r1 weak-curve audit: exact pre-admission diagnostic

crypto research harness

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Status

PRE-ADMISSION DIAGNOSTIC COMPLETE — NO ADMITTED SCIENTIFIC RUN OR INDEPENDENT IMPLEMENTATION VALIDATION EXISTS. A native diagnostic emitted deterministic exact certificates and replayed them with the same implementation. Its machine-readable artifact is [diagnostics/pre-admission-certificate.json](#), with embedded semantic certificate SHA-256 1cf4a3733e98c6acdc2219c24f29c15771fc67578bdb33dd426ab643e12572d8. The artifact index separately records the raw JSON byte SHA-256. The artifact explicitly records admitted_scientific_run=false and evidence_status=PRE_ADMISSION_DIAGNOSTIC_EXACT_CERTIFICATES. Its implemented_diagnostic_gates_passed=true field means only that the bounded gates implemented by this diagnostic replayed; it is not an admission, coverage, exploitability, or independent-validation verdict.

This is a completed **diagnostic**, not a completed experiment. The protocol and admission requirements remain in [README.md](#). Exact endpoint records and ordered routes are recorded in [degree5_curve_records.json](#) and [isogeny-routes.json](#). Reproduction commands, artifact hashes, toolchain details, and validation receipts are indexed in [ARTIFACT_INDEX.md](#). The editable diagram is [evidence-flow.dot](#), with rendered [SVG](#) and [PDF](#) views.

The diagnostic supports one unsurprising class-wide result, two exact transfer controls, and one conditional implementation result:

- At a frozen 80-bit classical threshold, the exactly certified prime subgroup has 56.325748 bits of modeled generic collision work (55.825379 with the stated negation optimization). The diagnostic verdict is `CLASS_WEAK`. This is a legacy parameter-size classification, independently corroborated by the published 2015 `sect113r1` discrete-log computation. It is not a new isogeny-induced break.
- Exactly two rational degree-5 kernels, codomains, full-coordinate map implementations, order-n images, forward planted-target homomorphisms, duals, and dual composition on both `G` and the planted target replayed. Both routes are `TRANSFER_ONLY_SPEEDUP_NOT_ESTABLISHED`; no codomain DLP was solved and no representative-specific attack or measured performance advantage was supplied.
- The repository's low-level binary point formulas do not use `b` during addition/doubling. One actual unchecked multiplier call on the singular same-`a`, `b=0` cubic was followed by exact BSGS/CRT recovery of the bounded planted scalar. This supports `CONDITIONAL_IMPLEMENTATION_WEAK` only if an attacker can reach that low-level API with a chosen point and distinguish its full-point output. No production protocol exposure was established. The singular cubic is not an elliptic curve and is not isogenous to `sect113r1`.

Evidence vocabulary

- **EXACT DIAGNOSTIC:** deterministic algebraic equality, certificate, or replay emitted and checked locally before admission.
- **MODELED:** analytic work estimate; not an executed attack or timing sample.
- **PRIOR ART:** a claim made by a cited external source, not measured here.
- **CONDITIONAL:** depends on an interface exposure not established here.
- **ADMITTED / INDEPENDENTLY VALIDATED:** absent. The verifier replays the artifact through the same native implementation and is not an independent implementation.

The diagnostic can establish exact equalities within its implementation, but it cannot promote them to admitted or independently reproduced scientific evidence. No wall-clock performance result, randomized rho run, prevalence estimate, or independent verdict is reported.

Exact source object

The source curve is SECG `sect113r1`:

$$E/\mathbb{F}_{2^{113}} : y^2 + xy = x^3 + ax^2 + b,$$

with polynomial basis modulus $z^{113} + z^9 + 1$ and SEC 2 coefficients:

```
a = 003088250CA6E7C7FE649CE85820F7
b = 00E8BEE4D3E2260744188BE0E9C723
Gx = 009D73616F35F4AB1407D73562C10F
Gy = 00A52830277958EE84D1315ED31886
n = 010000000000000000D9CCEC8A39E56F
h = 2
```

The source is [SEC 2 version 1.0](#), §3.2.1. The frozen PDF SHA-256 is:

`d1b16728ad83888fd656d16b99dc71bcd5541d42d848ffd0de7c62c19010d8c3`

sect113r1 weak-curve audit — exact pre-admission diagnostic outcome
 NOT AN ADMITTED SCIENTIFIC RUN · semantic certificate SHA-256 1c4a373e98c6acdc2219c24f29c1571fc67578bdb33dd426ab643e12572d8

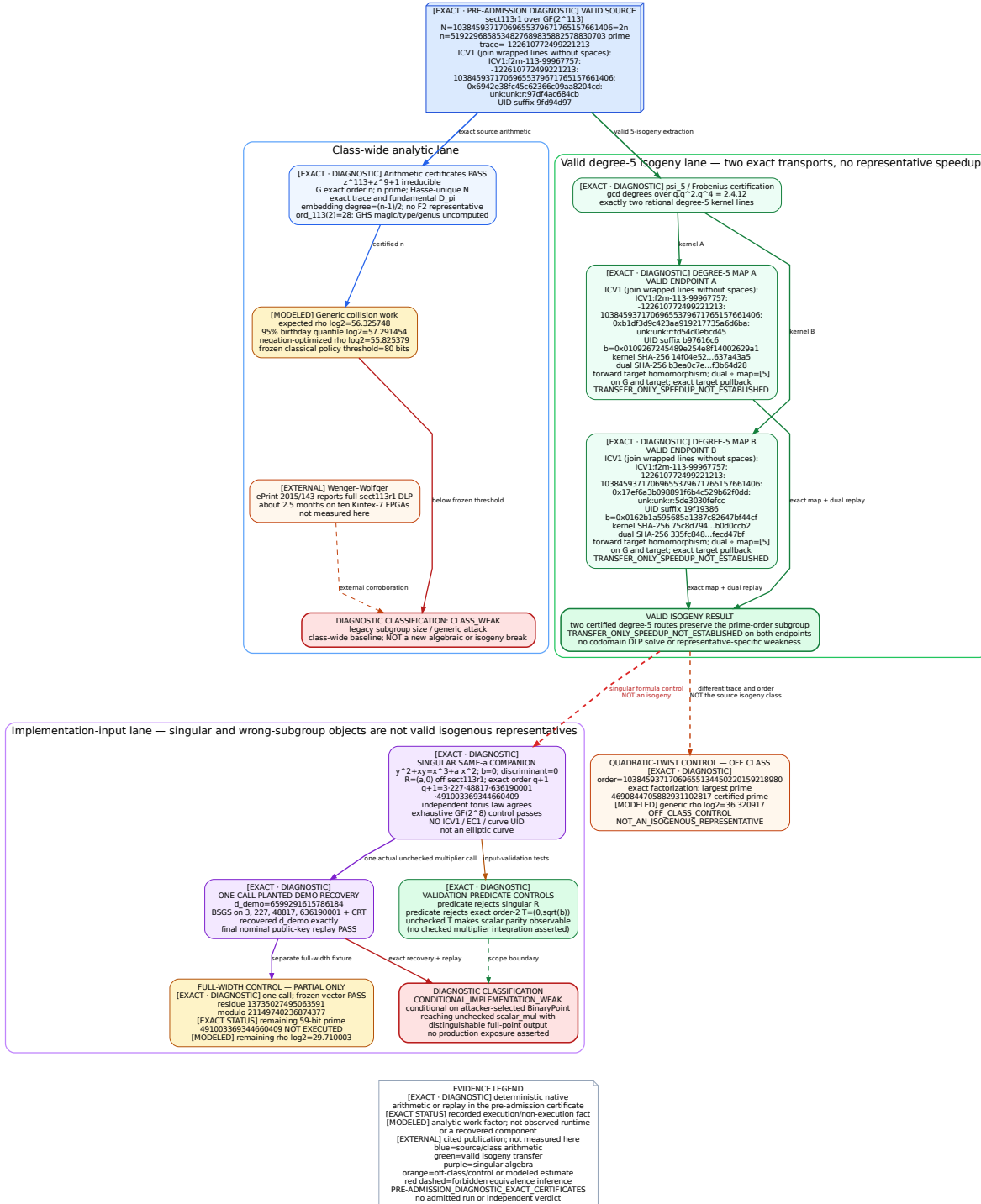


Figure 1: Evidence lanes for the exact source-class result, valid degree-5 transfer, and implementation-input findings. Solid green edges are exact certificate obligations; dashed amber edges remain modeled or conditional.

Resolved source representation identity:

```
icv1-f2m113-tm122610772499221213-97df4ac6
ICV1 (join wrapped lines without spaces):
ICV1:f2m-113-99967757:
-122610772499221213:
10384593717069655379671765157661406:
0x6942e38fc45c62366c09aa8204cd:
unk:unk:r:97df4ac684cb
EC1N113Csect113r1hf529f17bd191
curve UID (join wrapped lines without spaces):
urn:ec-record:1:sha256:
f529f17bd1913792333a661e3557ad6b8e0ca2d4d02b939bc069d17d9fd94d97
```

The diagnostic resolved these source strings. Each valid codomain has a different representation identity in [degree5_curve_records.json](#), and the ordered maps are bound in [isogeny-routes.json](#). The twist has a different curve identity. The singular control receives no curve identity.

Exact diagnostic source/class certificates

Order and generic security

The diagnostic exactly recomputed:

```
q = 2^113
  = 10384593717069655257060992658440192
n = 5192296858534827689835882578830703
N = #E(F_q) = 2*n
  = 10384593717069655379671765157661406
t = q + 1 - N = -122610772499221213
```

The native diagnostic verified a complete $n-1$ primality certificate for n , the generator's exact order, and the unique Hasse-interval multiple supporting $\#E(F_q)=2n$. It also verified irreducibility of $z^{113}+z^9+1$. These checks were replayed by the same implementation, not an independent point counter or validator. The generic references below are formulas, not benchmarks:

reference	modeled log2 group operations
$\sqrt{\pi n/2}$ expected collision work	56.32574806473616
$\sqrt{2n \ln(20)}$ 95% birthday quantile	57.29145434846796
frozen policy threshold	80

Thus the diagnostic classification is CLASS_WEAK. Because a rational isogeny preserves the trace and point count, the size classification applies to every valid representative in this class. A representative can be worse for a special attack, but it cannot restore more than the class's generic baseline.

PRIOR ART. Wenger and Wolfger state that they computed a full discrete logarithm on sect113r1 with ten Kintex-7 FPGAs, with their design processing 900 million iterations per second; see [ePrint 2015/143](#). This report will not rebrand those published observations as measurements from the present run.

Frobenius and endomorphism order

Exact diagnostic arithmetic:

```
D_pi = t^2 - 4*q
      = -26504973335422840129609279124569399
      = -7 * 47 * 411643769 * 195708607903277035369399
```

The diagnostic's exact factor/primality certificates make D_{π} squarefree and fundamental. Thus $Z[\pi]$ is maximal and the diagnostic found no conductor levels to hunt in the class. The minimum exact norm of a non-integer endomorphism is

$$(\text{abs}(D_{\pi})+1)/4 = 6626243333855710032402319781142350 \\ = \text{approximately } 2^{112.35181831539929}.$$

This excludes a cheap non-scalar CM/GLV endomorphism under the checked model. It does not exclude unmodeled algorithms.

Structural attack ledger

Attack family	Exact diagnostic result	Evidence boundary
Pohlig-Hellman	n prime with complete n-1 certificate	exact in diagnostic; no independent replay
MOV/Frey-Rück	$\text{ord}_n(q)=(n-1)/2$ exactly	exact in diagnostic; no independent replay
anomalous supersingular subfield/GLS	$N \neq q$ ordinary; t is odd 113 is prime; no degree-113 trace from an F_2 curve equals t	exact arithmetic exact arithmetic exact recurrence in diagnostic
GHS/Hess	$\text{ord}_{113}(2)=28$	exact orbit parameter only; magic number, type, genus, and applicability were not computed, so no attack conclusion
CM/GLV	fundamental D_{π} ; no low-norm non-scalar element	exact in diagnostic; scoped to checked model
direct Semaev/index calculus cover/Jacobian/extension transfer	no end-to-end result generated catalog contains same-field degree-3 $H \rightarrow E$ cover certificates for the source and both endpoints	unavailable; no negative claim cover existence only; subgroup transfer, inverse/recovery, end-to-end cost, and DLP advantage are untested ($\text{dlp_advantage}=\text{null}$)

No row may be converted from “not demonstrated” to “impossible” without the corresponding proof. In particular, equation counts or asymptotic discussion do not establish an executable index-calculus attack.

Valid-isogeny diagnostic

Modulo 5, the exact Frobenius polynomial factors as

$$X^2 - tX + q = (X - 3)(X - 4) \pmod{5}.$$

The diagnostic derived exactly two rational cyclic order-5 kernels from the fifth division polynomial rather than accepting expected coefficients:

$$\begin{aligned} Xq &= x^q \bmod \psi_5 \\ h1 &= \gcd(\psi_5, Xq + x) \\ Xq2 &= Xq^q \bmod \psi_5 \\ h2 &= \gcd(\psi_5, Xq2 + x) / h1 \end{aligned}$$

Both degree-2 factors passed squarefreeness, divisibility, Frobenius, and point-order checks. Each route produced a nonsingular Vélú codomain, full-point map, order-n image of G, and separately constructed dual. The diagnostic checked $\text{dual}(\phi(G)) = [5]G$, $\text{dual}(\phi([d]G)) = [5d]G$, and recovered the original planted target point as $[5^{-1} \bmod n] \text{dual}(\phi([d]G))$. This is an exact homomorphism and point-pullback control; it is not a discrete-log solve on the codomain. The native certificate recomputes each full ICV1/EC1/UID from the codomain model and generator. The endpoint and route JSON files are derived views cross-referenced to that certificate; their byte hashes and the source revision are bound by the artifact index, not by a separate native bundle verifier.

The diagnostic evaluated dual composition on G and the planted target, not on the broader protocol set of kernel points, infinity, and sign partners. Kernel/formula certificates and those two subgroup checks are strong transfer evidence, but they do not discharge that broader full-morphism validation obligation for a later admitted run.

Any future speedup claim must still charge the route cost

$$C^* = C_{\text{construct}} + C_{\text{map in}} + C_{\text{attack}} + C_{\text{map out}} + C_{\text{recover}} + C_{\text{verify}}.$$

No component may be silently treated as free. The diagnostic did not execute a matched source/codomain attack or collect timing samples, so it makes no performance claim.

Exact diagnostic route table:

Endpoint A has $b=0109267245489e254e8f14002629a1$; endpoint B has $b=0162b1a595685a1387c82647bf44cf$. Their complete identities and kernel digests are in the linked record and route JSON files.

route	valid curve/map certificate	subgroup replay	result
source	source order and generator	exact	incumbent; no timing claim
endpoint A	forward/dual kernels and formulas	order n; target composition and pullback; broader point-set check pending	transfer only; speedup not established
endpoint B	forward/dual kernels and formulas	order n; target composition and pullback; broader point-set check pending	transfer only; speedup not established

These are successful transfer controls, not new weak curves. The diagnostic found no representative-specific advantage in the tested routes, but it did not exhaust the isogeny class and cannot assert that no such representative exists.

Twist safety experiment

The exact diagnostic twist order is

10384593717069655134450220159218980
 $= 2^2 * 5 * 11 * 17 * 449 * 883 * 1493 * 4690844705882931102817$.

The diagnostic certified the factorization and primality of the largest factor 4690844705882931102817. Its generic collision work of about $2^{36.32092}$ is a model, not an executed solve. The twist is a different trace class and is classified OFF_CLASS_CONTROL_NOT_AN_ISOGENOUS_REPRESENTATIVE; no interface exposure was established.

Singular-input experiment

Algebraic object and cost

The deliberate control

$$S : y^2 + xy = x^3 + ax^2$$

has $b=0$ and discriminant zero. It is a singular cubic, not an elliptic curve. The exact diagnostic found $\text{Tr}(a)=1$, making the smooth locus a nonsplit torus with order

$$\begin{aligned} q+1 &= 10384593717069655257060992658440193 \\ &= 3 * 227 * 48817 * 636190001 * 491003369344660409. \end{aligned}$$

The point $P=(a,0)$ passed $[q+1]P=0$ and $[(q+1)/r]P \neq 0$ for every certified prime factor r . The largest factor has about 58.769 bits; its generic reference is about $2^{29.710}$ expected work, 26.616 bits below the nominal subgroup's modeled generic reference.

That comparison is an algebraic work model, not a wall-clock speedup. Any future end-to-end performance claim must count the projection, each discrete-log solve, CRT, transcript operations, and final replay.

Fixed public fixtures

The small-factor product is

$$M = 3 * 227 * 48817 * 636190001 = 21149740236874377.$$

The bounded proof fixture is:

```
domain = "crypto/sect113r1-singular/demo/v1\0"
d_demo = 1 + 0S2IP_BE(SHA256(domain)) mod (floor(M/2)-1)
d_demo = 6599291615786184 = 0x177205508734c8
```

Native BSGS on all four factors plus CRT recovered d_{demo} exactly from one actual call to the unchecked multiplier. The recovered residues were:

factor	residue	BSGS giant steps	exact replay
3	0	1	yes
227	58	4	yes
48817	26624	121	yes
636190001	487863039	19342	yes

CRT returned 6599291615786184, and the final nominal public-key replay matched. The expected scalar was not used by the solver path. The same native implementation replayed each equation; an independent implementation has not yet done so.

The full-width fixture is:

```
seed ASCII = "SECT113R1-SINGULAR-V1"
domain = "crypto/sect113r1-singular-recovery/seed/v1\0"
SHA256(domain || seed) = c4cfd5aa9d48d2079788cbdd3c6c7f579cf7d41cd002106eb2bab540ad639f08
d_full = 1 + 0S2IP_BE(SHA256(domain || seed)) mod (n-1)
d_full = 1108386698526968361534713354743651
      = 0x36a5ce943bb0bf6c27642c5b6f63
Qx = 0x00AF667D2D64DF714FCB7229D4509F
Qy = 0x0071C688F4F136AAF006193D1DC24F
Ux = 0x0078AED9A682BE670A024A24D76B1D
Uy = 0x00D567FA101A19AF65CE7F58C0D4A4
```

The public key Q and unchecked singular result U are post-search replay controls. They must not be used to prune BSGS or future rho candidates.

For the full-width fixture, one unchecked call produced the frozen output and the four executed components recovered $d_{full} \bmod M = 13735027495063591$. The remaining prime is 491003369344660409 ; its solve was explicitly **not executed**. The reported 29.710003-bit rho work is modeled. This is residue recovery, not full secret recovery.

Implementation boundary and controls

The checked code-level issue is that low-level binary addition, doubling, and scalar multiplication depend on a but do not validate b or subgroup membership. The diagnostic supports `CONDITIONAL_IMPLEMENTATION_WEAK` at this low-level surface because:

- the actual low-level multiplier accepted P ;
- its result agreed with a separate torus-law implementation, including an exhaustive $GF(2^8)$ control;
- the bounded scalar and the declared full-width residue were exactly replayed;
- the nominal equation and checked public-point predicate rejected P ;
- the predicate also rejected infinity and points outside the order- n subgroup; and
- caller reachability and a distinguishing production-protocol output remain explicit exposure obligations.

No result here licenses “weak isogenous curve.” The strongest result without a reachable wrapper is “conditional implementation weakness at the tested low-level surface.”

Nominal wrong-subgroup control

The on-curve point

$T = (0, 0x01DF93AA20579D377F97203F385B11)$

has exact order 2 in the diagnostic: $[2]T=0$, $[n]T=T$, and the subgroup predicate rejected it. Raw multiplication distinguished scalar parity. This is a standard unchecked-surface/wrong-subgroup control on the nominal curve, not evidence that the prime-order subgroup is easier; production exposure was not established.

How confidence will be earned

Exact algebra dominates this study. The diagnostic gains implementation-level confidence from machine-checkable certificates, mutation tests, a separate torus law, and final point replay. It does **not** have source-independent implementation validation: producer and replay verifier share the native code base. Repeating deterministic arithmetic does not create statistical confidence if both paths share a bug.

For any performance comparison, the admitted design must pin hardware and software, use isolated cores, randomize paired execution order, retain warmups and all failures, include an A/A check, and report every repetition plus a bootstrap interval on the paired effect. A speedup must use the same operation boundary and include construction and recovery.

For randomized rho, seeds and censoring budgets are frozen in advance. Success counts receive exact binomial intervals; time-to-hit uses survival analysis. The analytic birthday quantiles are not empirical confidence intervals.

For an isogeny walk, adjacent samples are dependent. Without a mixing theorem or defensible effective sample size, the report will state only the exact visited set, route coverage, and sample yield. A zero-hit walk is not evidence that the whole class contains no special representative.

Diagnostic verdict matrix

Object / claim	Diagnostic result	Boundary
prime subgroup at frozen 80-bit threshold	class weak; exact n, modeled 56.325748-bit rho work	legacy size; prior full DLP is external
two explicit valid degree-5 neighbors	exact kernels, maps, order-n images, duals, forward target homomorphism, and point pullback	same-implementation diagnostic; no codomain DLP solve
cheaper attack through a valid neighbor	not demonstrated; transfer only	no matched attack or timing study
singular same-a cubic accepted by unchecked multiplier	bounded scalar exactly recovered; full-width residue recovered	conditional implementation weakness at the tested low-level surface; attacker reachability/output remain obligations
full-width singular recovery	not completed	59-bit component unexecuted; rho cost modeled
public-point validation predicate rejects invalid/wrong-subgroup points	exact diagnostic control holds	no checked multiplier or production wrapper integration asserted
entire class has no other special attack	no claim	finite degree-5 controls are not exhaustive

The canonical scoreboard and progress timeline were checked but not updated: this diagnostic is neither an admitted IC/performance result nor a benchmark measurement. The registry-driven IC leaderboard and lab-browser data were regenerated because the two exact endpoint identities expand their curve roster. The study-local evidence-flow diagram was updated to show exact, modeled, external-prior-art, and conditional lanes.

References

1. Standards for Efficient Cryptography Group, [SEC 2 version 1.0](#), §3.2.1.
2. E. Wenger and P. Wolfger, "[Harder, Better, Faster, Stronger — Elliptic Curve Discrete Logarithm Computations on FPGAs](#)," [ePrint 2015/143](#).
3. P. Gaudry, F. Hess, and N. Smart, "Constructive and destructive facets of Weil descent on elliptic curves," *Journal of Cryptology* 15 (2002).
4. F. Hess, "Generalising the GHS attack on the elliptic curve discrete logarithm problem," *LMS Journal of Computation and Mathematics* 7 (2004).
5. A. Menezes and E. Teske, "Cryptographic implications of Hess' generalized GHS attack," *Applicable Algebra in Engineering, Communication and Computing* 16 (2006).
6. J. Vélú, "Isogénies entre courbes elliptiques," *Comptes rendus de l'Académie des sciences* 273 (1971).
7. J. Tate, "Endomorphisms of abelian varieties over finite fields," *Inventiones Mathematicae* 2 (1966).

External sources establish parameters, prior art, and theorem context. They do not replace the exact run certificates, explicit routes, exposure evidence, or matched end-to-end costs required above.